

Advanced Lesson 13 - Exercises:

Trace and Held Evaluation

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MATH 7013 - Advanced Mathematical Modeling

1: Tracing differentiation and integration

```
Clear[x];  
Trace[D[x^2]]  
Trace[D[(1 + x)^2]]  
Trace[D[(x + 1)^2]]  
Trace[Integrate[x + x^2 + e^x, x]]  
Trace[D[Integrate[(1 + x)^2, x]]]
```

What the traces show. Trace returns a nested list of the intermediate expressions Mathematica passed through, with the nesting recording how deep in the evaluation each step occurred. By default it reports only the steps that change something, which is why these traces are so much shorter than the calculations they describe - D and Integrate do their work inside built-in code that Trace does not open up.

The first trace has two entries: the input $D[x^2, x]$ and the result $2x$.

The second, for $(1 + x)^2$, is the same shape - input and result $2(1 + x)$.

The third, for $(x + 1)^2$, is the interesting one. It has an extra step at the very front: $x + 1$ becoming $1 + x$. Plus is Orderless, so the argument was sorted into canonical order before D was ever reached, and after that first line the trace is identical to the second. The way a sum is typed survives exactly one step and then vanishes.

The fourth trace shows the same thing again - $x + x^2 + E^x$ is reordered to $E^x + x + x^2$ first - and then the integral appears in one step as $E^x + x^2/2 + x^3/3$. The linearity is not visible, because it happens inside Integrate rather than as separate evaluation steps.

The fifth is the most instructive. The trace shows Integrate[(1 + x)^2, x] evaluated all the way to $x + x^2 + x^3/3$, and only then D applied to that result, giving $1 + 2x + x^2$. Mathematica does not notice that differentiation undoes integration and short-circuit the pair; it evaluates innermost first, because that is what ordinary evaluation order requires. The final answer is the expanded polynomial rather than the original $(1 + x)^2$, since the intermediate antiderivative came back expanded.

2: Tracing immediate against delayed replacement rules

```
Clear[a, b];
Trace[Sin[a] /. a -> b]
```

Part 1. With neither symbol assigned, the trace is almost empty: the rule matches, a becomes b, and Sin[b] is returned. Nothing else happens.

```
Clear[a, b];
b = 5;
Trace[Sin[a] /. a -> b]
```

Part 2. The rule $a \rightarrow b$ is an immediate rule, so its right-hand side is evaluated before the replacement is attempted. The trace shows b evaluating to 5 first, so the rule that actually gets applied is $a \rightarrow 5$, and the result is Sin[5].

```
Clear[a, b];
a = 2;
b = 5;
Trace[Sin[a] /. a -> b]
```

Part 3. Now both sides evaluate before ReplaceAll runs. The expression Sin[a] becomes Sin[2] and the rule becomes $2 \rightarrow 5$, which still matches, so the answer is Sin[5]. The trace shows both evaluations happening ahead of the replacement. Note the accident: it only worked because the rule's left-hand side evaluated to the same thing the expression did.

```
Clear[a, b];
Trace[Sin[a] /. a -> b]
```

Part 4. With nothing assigned, RuleDelayed behaves exactly like Rule and the trace is the same. The difference between them is invisible until the right-hand side has something to evaluate.

```
Clear[a, b];
b = 5;
Trace[Sin[a] /. a -> b]
```

Part 5. Here the two diverge. RuleDelayed holds its right-hand side, so b is not evaluated when the rule is constructed. It is substituted as the symbol b, and only then, as part of the ordinary evaluation of the result, does b yield 5. The final answer is again Sin[5], but the trace reaches it by a different route - the evaluation of b appears after the replacement instead of before it.

```
Clear[a, b];
a = 2;
b = 5;
Trace[Sin[a] /. a -> b]
```

Part 6. The result is Sin[5] once more.

The observation the exercise asks for. The left-hand side of a delayed rule is not held: a is evaluated to 2 in the rule just as it is in the expression. But that makes no difference to the answer, because it happens on both sides at once - the expression becomes Sin[2] and the rule becomes 2 -> b, and they still match. Only the right-hand side is protected by RuleDelayed, and only the right-hand side needs to be, since that is where the substituted material comes from.

3: Displaying arguments unevaluated and evaluated, side by side

```
ClearAll[f, a];
SetAttributes[f, HoldAll];
f[args___] := List @@ Function[u, {HoldForm[u], u}, HoldAll] /@ Hold[args]
```

How this works, since the exercise offers no hints.

The function must see its arguments before they are evaluated, so it is given the HoldAll attribute. Inside, the held sequence is captured by a BlankNullSequence and immediately wrapped: Hold[args] is a Hold expression containing all of the arguments, still unevaluated.

Map applies a function to each element of that Hold at level one. The function being mapped is itself a pure function declared HoldAll, which is the essential trick - a plain pure function would evaluate its argument on the way in and the unevaluated form would be lost before anything could record it. Note that this is Function with a named parameter and a third argument giving attributes, not the ampersand form; the ampersand form has

no place to put HoldAll.

For each argument the mapped function builds a two-element list: HoldForm[u], which keeps the argument in its original form for display, and a bare u, which is evaluated in the ordinary way as the body evaluates. One argument, two renderings.

Finally Apply replaces the outer Hold with List, releasing the structure into an ordinary list of pairs.

```
Clear[a];
f[N[ $\pi$ ], 1 + 2, a a2, a (1 + a)]
TableForm[f[N[ $\pi$ ], 1 + 2, a a2, a (1 + a)]]
```

The result is

```
{{N[Pi], 3.141592653589793}, {1 + 2, 3}, {a a^2, a^3}, {a (1 + a), a (1 + a)}}
```

matching the target output in the exercise, and TableForm lays it out as four rows of two columns.

The fourth row is the one that shows the mechanism is honest: $a (1 + a)$ appears identically in both columns, because Mathematica does not expand products of sums on its own. It is not that the evaluated column failed - that expression really is its own evaluated form. The third row is the contrast: $a a^2$ is collected into a^3 by the automatic rules for Times, so the two columns differ.